

Predicting Eruptive Events at Kilauea from Earthquake Data

Reproduction of Mullet, Singh and Wong (Stanford CS229, 2019), extended with an RBF Support Vector Machine
UE24CS352A Machine Learning - Mini-Project Write-up | Team: [Member 1 name, SRN], [Member 2 name, SRN]

1. Problem statement

The reference paper asks whether the earthquake catalog alone can tell us about eruptions at Kilauea volcano (Hawaii). It defines two tasks: **(a) contemporaneous eruption classification**, predicting for each earthquake whether the volcano is erupting (1) or in repose (0), and **(b) time-to-eruption regression**, predicting the days until the next eruption starts (0 if one is under way). Our project has two parts: **Part 1** reproduces the paper's pipeline and compares results with the reported ones; **Part 2** adds one model the paper does not use (an RBF SVM) and evaluates it on the same split and metrics.

2. Dataset

- **Catalog:** ANSS earthquake catalog from the USGS ComCat API, box longitude -155.5 to -155.0, latitude 19.1 to 19.5 (East Rift Zone near Pu'u 'O'o, read from the paper's Figure 1).
- **Size:** the paper does not state its date range or magnitude cutoff. We used 2000-2016 and chose the cutoff closest to the paper's 3,764 events on a coarse grid (magnitude 2.0), giving 6,207 events, or **5,950 after labeling**. Our catalog is therefore larger than the paper's.
- **Labels:** the paper's eruption table is not published, so we approximated three eruption episodes from HVO / Smithsonian reports (June 2007, March 2011, and June 2014 to May 2016). About 18.5% of earthquakes are labeled erupting (paper: about 20%).
- **Features (7):** longitude, latitude, depth, magnitude, and earthquake counts over the previous 1, 7 and 30 days.
- **Split:** stratified 70/20/10 train/dev/test (4,165 / 1,190 / 595 earthquakes, seed 42). Features and the regression target are standardized with training-set statistics only.

3. Approach

Part 1 models (as in the paper). Logistic regression (classification only); K-means (8 clusters per class for classification, 10 clusters on features plus target for regression, nearest-centroid prediction); random forest (3 features per split; depth and number of trees tuned on dev); and a fully connected neural network (4 hidden layers of 1,000 ReLU units, SGD with momentum, batch size 50, weight decay, BCE or MSE loss, 100 epochs, learning rate and weight decay tuned on dev). **Metrics:** Cohen's kappa and AUROC for classification, RMSE and R^2 for regression. Because about 80% of labels are negative, the decision threshold (not stated in the paper) is chosen on the dev set to maximize kappa and then applied to train and test.

Part 2 model: RBF SVM (SVC for classification, SVR for regression). We chose it because it is not in the paper, handles both tasks, works on our standardized features, and gives a smooth non-linear boundary that contrasts with the random forest's axis-aligned splits and the neural network's roughly three million weights. C in $\{0.1, 1, 10, 100\}$ and γ in $\{\text{scale}, 0.01, 0.1, 1\}$ were tuned on dev (AUROC for SVC, R^2 for SVR). Selected: SVC with $C = 100$, $\gamma = \text{scale}$; SVR with $C = 1$, $\gamma = 1$. The SVC value of C and the SVR value of γ lie on the grid edge, so larger values were not tested.

4. Implementation overview

Everything is in one Python notebook (pandas, scikit-learn, PyTorch; run on a Colab GPU). The steps are: download the catalog year by year from the USGS API; filter by magnitude; label each earthquake from the eruption episodes; compute the 1/7/30-day rate features with a sorted-time search; split and scale; train each model; tune hyperparameters on dev only; and evaluate on train, dev and test with shared helper functions so every model is scored the same way. The test set is never used for tuning.

5. Results (test set)

Classification	Kappa (paper)	Kappa (ours)	AUROC (paper)	AUROC (ours)
Logistic regression	0.28	0.36	0.60	0.74
K-means	0.24	0.51	0.63	0.80
Random forest	0.52	0.59	0.71	0.91
Neural network	0.29	0.52	0.61	0.87
SVM (new, Part 2)	n/a	0.55	n/a	0.82

Regression	RMSE (paper)	RMSE (ours)	R2 (paper)	R2 (ours)
K-means	1.02	1.09	0.037	-0.111
Random forest	0.64	0.75	0.380	0.476
Neural network	0.90	0.77	0.190	0.444
SVM (new, Part 2)	n/a	0.80	n/a	0.407

Paper values are from its Figures 4 and 5 (test rows). For the RF classifier the paper's text says kappa 0.42 but its table says 0.52; we compare against the table. The paper reports no regression result for logistic regression and has no SVM results.

What matches. Random forest is the best classifier and the best regressor in both the paper and our runs, and K-means regression has no skill in either. **What differs.** All our classification scores are higher than the paper's (kappa by 0.07 to 0.27, AUROC by 0.14 to 0.26). The paper's weak models (K-means, neural network, logistic regression) are weak there, while in our run K-means and the NN come close to the random forest, so the ranking of weaker models is not reproduced. The random forest also overfits strongly (train kappa 0.999 vs dev 0.59), and the NN less so (train 0.64 vs test 0.52).

Likely reasons for the gaps.

- **Our approximate eruption labels.** All 2015-16 earthquakes are labeled erupting and most other years are 0%, so location and rate features can act as a proxy for the year. This is a likely cause of the larger AUROC, but we did not test it directly.
- **Random split.** Neighboring earthquakes with overlapping rate windows fall in both train and test, which inflates scores (the paper shares this design).
- **Different catalog and unreported settings.** 5,950 vs 3,764 events; the paper's threshold, NN learning rate, weight decay, epochs, and forest grid are unknown, so we tuned them on dev. NN results also vary slightly between runs (for example test kappa 0.52 vs 0.48).

6. Conclusions

- The paper's pipeline, split, features and metrics were reproduced and the paper's numbers are transcribed correctly. The **qualitative finding holds** (random forest is best, K-means regression fails), but our **absolute scores are higher** than the paper's, mainly because of our different catalog and approximate eruption labels. The results should be compared with each other rather than with the paper's absolute values.
- **The SVM** is second on kappa (0.55; RF 0.59, NN 0.52), third on AUROC (0.82; RF 0.91, NN 0.87) and third of four on regression (R^2 0.41; RF 0.48, NN 0.44). It clearly beats logistic regression and slightly exceeds K-means. It overfits less than the RF and NN (kappa 0.60 train to 0.55 test). A plausible reason for the RF's edge is that tree splits on location and rate separate our labels, including the 2014-16 block, better than a smooth kernel; we did not test this. The test set has only 110 erupting earthquakes, so differences of about 0.03 in kappa are within noise.
- **Future work:** use the real eruption table, a chronological split to test the year-leakage explanation, a wider SVM grid, and extra data (deformation, gas) as the paper suggests.